

SALDEV 1371 ARR Clone Data Integrity Architect Review

FlywirePartial | Read-only non-stepped quote-line cloning best-practice review

Source	SALDEV-1371 FlywirePartial Enterprise sandbox 00DhG0000000jOXUAY
Scope	2 QCP scripts non-stepped line/group clone ARR nulling
Review date	4 September 2026
Validation mode	Live script read-back field schema JS syntax contextual Apex evidence
Decision	NO-GO - Changes Requested before story closure
Boundary	Read-only review; no clone execution, record save, metadata edit, or deployment

ARCHITECT DECISION

Changes Requested. SALDEV-1371 cannot be closed from code presence alone. The clone script omits the ARR fields from its dependency declaration, Domestic_Sponsor__c is not a real quote-line field, the field list is duplicated across scripts, and no saved quote-line read-back proves the ARR values are null outside QLE.

Acceptance and quality outcome

#	Scope	Result	Evidence
1	Clone handler clears ARR fields	IMPLEMENTED	A 25-field nulling list runs in the clone handler; JavaScript syntax passes.
2	All field APIs are valid	FAIL	Domestic_Sponsor__c was not found on SBQQ__QuoteLine__c.
3	Runtime dependencies are complete	FAIL	The clone script declares only NS ID and Opportunity Line ID, not the ARR fields it writes.
4	Saved clone values are null	NOT PROVEN	No record-detail or SOQL read-back was performed for a cloned non-SUP line.
5	Line and group clones covered	MISSING	No functional matrix proves both non-stepped clone entry points.
6	SALDEV-1351 regression passes	MISSING	The APAC ARR Flow outcome was not re-tested after a clone save.

Clone path and missing acceptance proof

SALDEV-1371 requires ARR values to be null on saved cloned quote-line records for non-stepped quoting. Hiding fields in QLE is not acceptance evidence; database read-back and the SALDEV-1351 regression outcome are required.

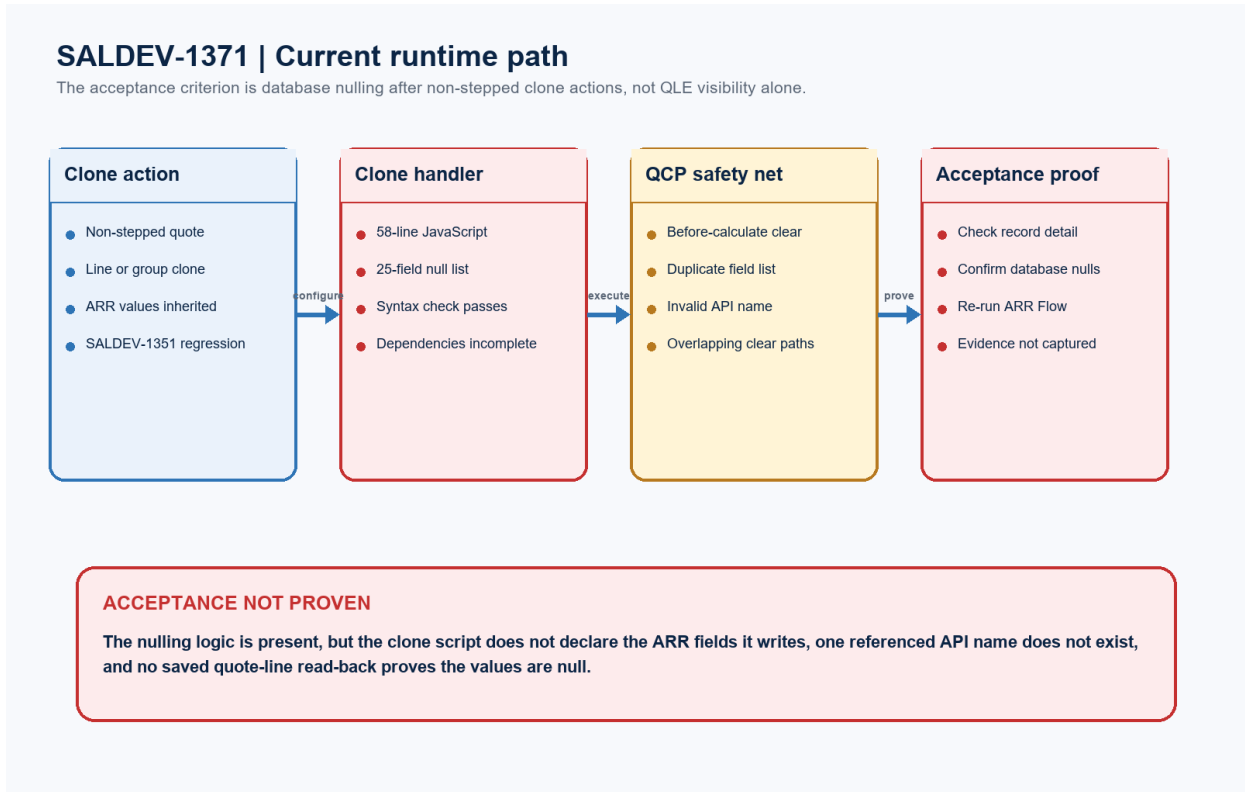


Figure 1. SALDEV-1371 clone path and evidence gap

DESIGN READING

Blue shows the non-stepped clone acceptance path. **Green** shows code characteristics worth retaining. **Amber** marks incomplete evidence. **Red** identifies schema or proof defects that block story closure.

Detailed code and validation evidence

1. Story and live runtime evidence

Evidence area	Observed result	Evidence
Story status	Duplicate	SALDEV-1371 states the requirement is handled with SALDEV-1427.
Target org	Verified sandbox	FlywirePartial is org 00DhG0000000jOXUAY; Enterprise Edition; IsSandbox=true.
QLE_CloneLineHandler	Runtime present	58 JavaScript lines; clone hook and runtime exist; syntax check passed.
QCP_HideFieldInQLE	Safety net present	The before-calculate path repeats the ARR nulling list; syntax check passed.
Field read-back	24 of 25 found	Domestic_Sponsor__c does not exist; the approved Processed Volume Amount exclusion is preserved.

2. Best-practice findings

Finding	Severity	Evidence
Dependency contract	BLOCKER	The clone script writes ARR fields that are absent from SBQQ_QuoteLineFields__c.
Schema correctness	BLOCKER	Domestic Sponsor percent is named in the story, but Domestic_Sponsor__c is not in the live schema.
Maintainability	HIGH	The ARR list is duplicated across scripts and the clone handler has overlapping record-clearing paths.

3. Acceptance evidence and boundaries

Test / boundary	Status	Evidence
JavaScript syntax	PASS	Both live scripts parse successfully with Node syntax checks.
Focused Apex context	PASS	7 of 7 handler tests passed, but they do not execute the JavaScript clone hook.
Apex dry-run context	FAILED	Coverage failed at 43.59% for QuoteLineTrigger; this is not proof of ARR nulling.
JavaScript unit proof	MISSING	No automated test demonstrates each ARR field changes from populated to null.
Record-level acceptance	MISSING	No saved non-SUP clone was checked outside QLE or against the ARR Flow outcome.

Required clone-data architecture and proof model

A release-ready solution should use one verified ARR schema, clear values once at clone time, surface failures, and prove every saved quote-line field outside QLE for both line and group clones.

SALDEV-1371 | Required release design

The clone contract must be schema-valid and proven on saved quote-line records

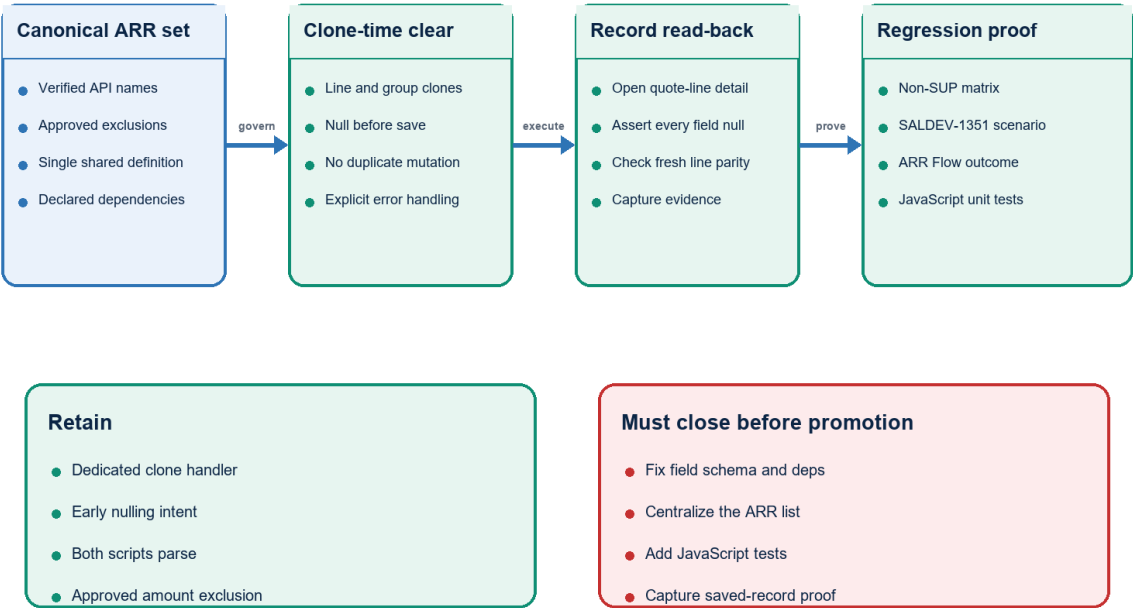


Figure 2. SALDEV-1371 required clone contract and record-level proof

Architect interpretation: the nulling intent is visible and both scripts parse, but a valid runtime field contract and saved-record proof are mandatory. The Apex dry run is contextual release evidence and does not validate this JavaScript acceptance criterion.

Required completion gates

Gate	Required action	Owner	Evidence needed
1	Correct the ARR schema	CPQ Developer	Map Domestic Sponsor percent to a verified API or remove the invalid name.
2	Declare every ARR dependency	CPQ Developer	Clone and calculate scripts expose every field they read or write.
3	Centralize nulling logic	CPQ Developer	One canonical field definition and one clear mutation path with visible errors.
4	Add JavaScript unit tests	Developer	Populated-to-null assertions for every ARR field and both clone entry points.
5	Run non-SUP functional matrix	CPQ QA	Line clone and group clone records show null ARR values outside QLE.
6	Re-test SALDEV-1351 outcome	CPQ QA / RevOps	Saved clone does not inflate ARR and the background Flow produces the expected result.

Ready-to-paste architect response

STORY COMMENT

Architect review completed for SALDEV-1371 in FlywirePartial. Decision: Changes Requested / NO-GO for story closure. The live QLE_CloneLineHandler contains a 25-field nulling list and both related QCP scripts pass JavaScript syntax checks. However, the clone script declares only NS ID and Opportunity Line ID rather than the ARR fields it writes; Domestic_Sponsor__c is not present on SBQQ__QuoteLine__c; the ARR list is duplicated across scripts; and no JavaScript unit or saved-record evidence proves non-stepped line and group clones have null ARR values outside QLE. The 7 passing Apex tests and failed Apex check-only run are contextual only and do not exercise this JavaScript acceptance path. Correct the schema and dependencies, centralize the field contract, add exact JavaScript tests, verify cloned quote-line records outside QLE, and re-run the SALDEV-1351 ARR regression before closing the duplicate story. No deployment or Salesforce change occurred.

Evidence boundary

Evidence: the SALDEV-1371 source document, verified FlywirePartial sandbox and scripts, field-definition read-back, syntax checks, and contextual Apex validation. No cloned records were created or saved, and no Salesforce record or metadata changes, deployment, activation, promotion, or Production action occurred.